LECTURE 4

Compactness: Heine–Borel and Bolzano–Weierstrass

MATH S4061 | Introduction to Modern Analysis I

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Mathematics is the science of the infinite.

Hermann Weyl

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Overview. Lecture 3 defined compactness—every open cover has a finite subcover (3.4.1)—and showed it is intrinsic (3.4.4). Here we draw out its consequences: a compact set is closed and bounded, a closed subset of a compact set is compact, and nested nonempty compact sets meet. Specialising to $\mathbb{R}^k$ through nested intervals and k-cells yields the two pillars of the subject—*Heine–Borel* (in $\mathbb{R}^k$, compact = closed and bounded) and *Bolzano–Weierstrass* (every bounded infinite set has a cluster point). A short topological coda—interior, closure, connectedness—closes the chapter.

4.1 Compactness, closedness, and boundedness

Theorem 4.1.1

A compact set is closed and bounded

A compact subset K of a metric space (X, d) is closed and bounded (4.3.4).[1]

Proof 4.1.1

Direct Proof that the Complement Is Open.

(1) If $K = \emptyset$ it is closed and bounded vacuously, so assume $K \neq \emptyset$. Fix $p \notin K$. For each $q \in K$ set $r_q = \frac{1}{2}d(p, q) > 0$; the triangle inequality makes $B_{r_q}(p) \cap B_{r_q}(q) = \emptyset$.

(2) The balls $\{B_{r_q}(q)\}_{q \in K}$ cover K, so by compactness finitely many do: $K \subseteq B_{r_{q_1}}(q_1) \cup \cdots \cup B_{r_{q_n}}(q_n)$.

(3) Put $r = \min(r_{q_1}, \ldots, r_{q_n}) > 0$. Then $B_r(p)$ is disjoint from every $B_{r_{q_i}}(q_i)$, hence from K. So p is interior to K^c; as $p \notin K$ was arbitrary, K^c is open and K is closed. ■

[Direct] ball separation; cf. R 2.34.

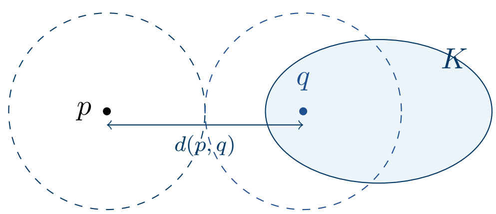

Figure 4.1.2. SEPARATING A POINT FROM A COMPACT SET. For $p \notin K$ and $q \in K$, the balls of radius $r_q = \frac{1}{2}d(p, q)$ about p and q are disjoint; as q ranges over K they cover it, and a finite subcover walls p off — so K^c is open (proof of 4.1.1).

Proposition 4.1.3

A closed subset of a compact set is compact

If K is compact and $F \subseteq K$ is closed in K, then F is compact.

Proof 4.1.3

Direct Proof that a Closed Subset of a Compact Set Is Compact.

(1) Let $\{V_\alpha\}$ be an open cover of F. Since F is closed, F^c is open in K, and $\{V_\alpha\} \cup \{F^c\}$ is an open cover of K.

(2) By compactness, finitely many cover K: $K \subseteq V_{\alpha_1} \cup \cdots \cup V_{\alpha_n} \cup F^c$.

(3) Since $F \cap F^c = \emptyset$, dropping F^c still covers F: $F \subseteq V_{\alpha_1} \cup \cdots \cup V_{\alpha_n}$. So every open

[1] Bounded: fix any p; the balls $B_n(p)$ $(n \geq 1)$ cover K, so finitely many do, and $K \subseteq B_N(p)$ for the largest N.

cover of F has a finite subcover, and F is compact.

■

[Direct] cf. R 2.35.

4.2 The finite-intersection property and nested sets

Proposition 4.2.1

The finite-intersection property

Let $\{K_\alpha\}$ be a family of nonempty compact subsets of X such that every finite subfamily has nonempty intersection. Then $\bigcap_\alpha K_\alpha \neq \emptyset$.

Proof 4.2.1

Proof by Contradiction of the Finite-Intersection Property.

(1) Suppose $\bigcap_\alpha K_\alpha = \emptyset$ and fix one K_1. Then $K_1 \cap \bigcap_{\alpha\neq 1} K_\alpha = \emptyset$, i.e. $K_1 \subseteq \left(\bigcap_{\alpha\neq 1} K_\alpha\right)^c = \bigcup_{\alpha\neq 1} K_\alpha^c$.

(2) Each K_α is compact, hence closed (4.1.1), so each K_α^c is open: $\{K_\alpha^c\}_{\alpha\neq 1}$ is an open cover of K_1.

(3) By compactness of K_1, finitely many cover it: $K_1 \subseteq K_{\alpha_1}^c \cup \cdots \cup K_{\alpha_n}^c$, i.e. $K_1 \cap K_{\alpha_1} \cap \cdots \cap K_{\alpha_n} = \emptyset$ — a finite subfamily with empty intersection, contradicting the hypothesis.

■

[A9] uses 4.1.1; cf. R 2.36.

Corollary 4.2.2

Nested nonempty compact sets meet

A nested sequence $K_1 \supseteq K_2 \supseteq \cdots$ of nonempty compact sets has $\bigcap_n K_n \neq \emptyset$.

Proposition 4.2.3

Nested closed intervals in $\mathbb{R}$

A nested sequence of nonempty closed intervals $I_n = [a_n, b_n]$ $(I_{n+1} \subseteq I_n)$ has $\bigcap_n I_n = [\sup_n a_n,\ \inf_n b_n] \neq \emptyset$.

Proof 4.2.3

Proof by a Supremum Argument that Nested Closed Intervals Meet.

(1) Nesting gives $a_1 \leq a_2 \leq \cdots \leq b_2 \leq b_1$, so every b_m is an upper bound for $\{a_n\}$. By the least-upper-bound property (1.6.1), $\alpha = \sup_n a_n$ exists and $a_n \leq \alpha \leq b_n$ for all n; thus $\alpha \in \bigcap_n I_n$.

(2) Moreover $x \in \bigcap_n I_n \iff a_n \leq x \leq b_n\ \forall n \iff \sup_n a_n \leq x \leq \inf_n b_n$, so $\bigcap_n I_n = [\sup_n a_n, \inf_n b_n]$.

■

[A5] uses 1.6.1; cf. R 2.38.

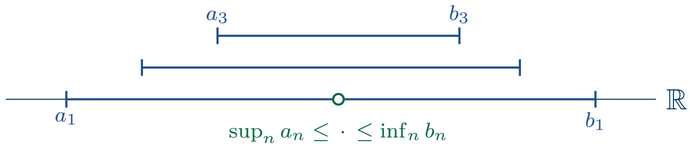

Figure 4.2.4. NESTED CLOSED INTERVALS. A descending chain $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$ shares every point of $[\sup_n a_n, \inf_n b_n]$ — in particular it is never empty (4.2.3).

Definition 4.2.5

k-cell

A *k-cell* (R 2.17) is a product of closed intervals

$$I = [a_1, b_1] \times [a_2, b_2] \times \cdots \times [a_k, b_k] \subseteq \mathbb{R}^k.$$

Proposition 4.2.6

Nested k-cells meet

A nested sequence of nonempty k-cells $I_1 \supseteq I_2 \supseteq \cdots$ has $\bigcap_n I_n \neq \emptyset$.

Proof 4.2.6

Proof Coordinatewise that Nested k-Cells Meet.

(1) Write $I_n = [a_1^{(n)}, b_1^{(n)}] \times \cdots \times [a_k^{(n)}, b_k^{(n)}]$. For each fixed coordinate i the intervals $[a_i^{(n)}, b_i^{(n)}]$ are nested, so by 4.2.3 some $\alpha_i \in \bigcap_n [a_i^{(n)}, b_i^{(n)}]$.

(2) Then $\alpha = (\alpha_1, \ldots, \alpha_k) \in \bigcap_n I_n$.

■

[A5] applies 4.2.3 in each coordinate; cf. R 2.39.

4.3 Heine–Borel and Bolzano–Weierstrass

Theorem 4.3.1

Every k-cell is compact

Every k-cell $I \subseteq \mathbb{R}^k$ is compact.

Proof 4.3.1

Proof by Repeated Bisection that the k-cell Is Compact.

(1) Let $\delta = \left(\sum_{i=1}^k (b_i - a_i)^2\right)^{1/2}$, so $d(x, y) \le \delta$ for all $x, y \in I$. Suppose I is *not* compact: some open cover $\{G_\alpha\}$ has no finite subcover.

(2) Bisecting each edge at its midpoint splits I into 2^k sub-cells. If each had a finite subcover, so would I; hence some sub-cell I_1 has none. Iterating gives nested cells $I \supseteq I_1 \supseteq I_2 \supseteq \cdots$, each without a finite subcover, with $\operatorname{diam} I_n \le 2^{-n}\delta$ (each

bisection halves every edge, hence the diagonal).

(3) By 4.2.6 some $x \in \bigcap_n I_n$. As $\{G_\alpha\}$ covers I, $x \in G_{\alpha_0}$ for some α_0, and G_{α_0} open gives $r > 0$ with $B_r(x) \subseteq G_{\alpha_0}$. Choose N with $2^{-N}\delta < r$ (Archimedean, 1.7.1); then $I_N \subseteq B_r(x) \subseteq G_{\alpha_0}$ — so I_N is covered by the single G_{α_0}, contradicting that it has no finite subcover.

■

[A7] bisection and nested cells (4.2.6); cf. R 2.40.

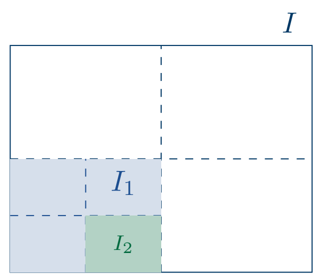

Figure 4.3.2. Bisecting a k-cell. A k-cell without a finite subcover has a 2^k-bisection inheriting the defect; iterating gives nested cells $I \supset I_1 \supset I_2$ of diameter $\to 0$, trapping a point that a single cover element must contain (proof of 4.3.1).

Theorem 4.3.3

Bolzano–Weierstrass

Every infinite subset E of a compact metric space[2] K has a cluster point in K.

Proof 4.3.3

Proof by Contradiction that E Would Be Finite.

(1) Suppose no point of K is a cluster point of E. Then each $q \in K$ has $r_q > 0$ with $B_{r_q}(q) \cap E \subseteq \{q\}$.

(2) The balls $\{B_{r_q}(q)\}_{q\in K}$ cover K; by compactness finitely many do. Each holds at most one point of E, so E is finite — contradicting that E is infinite.

■

[A9] cf. R 2.37.

Definition 4.3.4

Bounded set

$E \subseteq X$ is *bounded* (R 2.18) if $E \subseteq B_r(p)$ for some $p \in X$ and $r > 0$.

Theorem 4.3.5

Heine–Borel

[2] Rudin reserves the name *Bolzano–Weierstrass* (R 2.42) for the bounded-$\mathbb{R}^k$ case; the general compact statement here is R 2.37, whence the $\mathbb{R}^k$ case follows by enclosing a bounded set in a k-cell (4.3.1).

For $E \subseteq \mathbb{R}^k$ the following are equivalent:[3]

1. E is closed and bounded;
2. E is compact;
3. every infinite subset of E has a cluster point in E.

Proof 4.3.5

Proof of the Heine–Borel Theorem by Cycling (a)⇒(b)⇒(c)⇒(a).

(1) (a)⇒(b). Bounded E lies in some k-cell I; I is compact (4.3.1) and $E \subseteq I$ is closed, so E is compact (4.1.3).

(2) (b)⇒(c). This is Bolzano–Weierstrass (4.3.3).

(3) (c)⇒(a), contrapositive. If E is *not closed*, a cluster point $q \notin E$ exists; choose $p_n \in E$ with $\|p_n - q\| < \frac{1}{n}$. The set $\{p_n\}$ is infinite (a finite range would repeat some value $p \in E$ infinitely, forcing $q = p \in E$), and its only possible cluster point is $q \notin E$: any $y \in E$ has $\|p_n - y\| \geq \|q - y\| - \frac{1}{n} \geq \frac{1}{2}\|q - y\|$ eventually. If E is *not bounded*, choose $p_n \in E$ with $\|p_n\| > n$; then $\{p_n\}$ is infinite (the norms are unbounded) with no cluster point, since any $B_1(y)$ holds only finitely many p_n. Either way (c) fails.

■

[A9+A4] uses 4.1.3, 4.3.1, 4.3.3; cf. R 2.41.

4.4 Interior, closure, and connectedness

Definition 4.4.1

Interior and closure

For $E \subseteq X$, the *interior* $\mathring{E}$ is the set of interior points of E (2.3.3); it is open, and E is open iff $E = \mathring{E}$. The *closure* (R 2.27) is

$$\overline{E} = E \cup \{\text{cluster points of } E\} \text{ (3.2.1);}$$

it is closed — indeed the smallest closed set containing E.[4]

Definition 4.4.2

Connected and disconnected

A metric space (X, d) is *connected* (R 2.45) if its only subsets that are both open and closed ("clopen") are $\emptyset$ and X. Equivalently, X is *disconnected* when $X = A \cup B$ for nonempty, disjoint, open sets A, B.

[3]The equivalence (a) $\iff$ (b) is *special to* $\mathbb{R}^k$: in a general metric space a compact set is closed and bounded (4.1.1), but closed-and-bounded need not be compact.

[4]If $C \supseteq E$ is closed then C contains every cluster point of E, so $C \supseteq \overline{E}$; and $\overline{E}$ is itself closed, so it is the least such set.

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